

Sidney Rafilson

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EDUCATION

New York University — *New York, NY*

Ph.D. student in Neural Science

September 2025 – Present (expected 2030)

University of Oregon — *Eugene, OR*

B.S. in Mathematics and Neuroscience

September 2020 - June 2024

PUBLICATIONS

- Paleologos, Vöröslakos, **Rafilson**, et al., (2026) *Reusable robotic microdrive for silicon neural probes*. In Review
- **Rafilson et al.** (2025) *Challenges in inferring breathing rhythms from olfactory bulb local field potentials*. Chemical Senses
- Sterrett, Findley, **Rafilson**, et al. (2025). *Olfactory bulb tracks breathing rhythms and place in freely behaving mice*. eLife

PRESENTATIONS

- **Rafilson et al.**, (2026). *Anatomical propagation organizes the temporal structure of hippocampal replay*. Poster presented at the Society for Neuroscience conference.
- **Rafilson et al.**, (2025). *Neural correlates of sniffing and place in simultaneous recordings from the olfactory bulb and hippocampus*. Poster presented at the Association for Chemoreception Sciences conference
- **Rafilson et al.**, (2024). *Challenges in inferring breathing rhythms from olfactory bulb local field potentials*. Poster presented at the Society for Neuroscience conference.
- **Rafilson et al.**, (2024). *Olfactory bulb local field potentials track breathing rhythms at multiple time scales*. Poster presented at the Association for Chemoreception Sciences conference.
- **Rafilson et al.**, (2020). *Internet Addiction and its Correlates*. Abstract accepted for poster presentation at the annual American Psychological Association conference.

RESEARCH EXPERIENCE

Ph.D. Student — Buzsáki Lab, New York University Langone School of Medicine
September 2025 – Present

- Surgical implantation of silicon probes in rats, behavioral and neural recordings, perfusion and histological verification.
- Developed spike-sorting and data alignment pipeline for multiple simultaneous Neuropixels recordings; software adopted widely within the laboratory.
- Developed a spike-sorting pipeline for human iEEG recordings.
- Characterizing propagation patterns of sharp-wave ripples along the septotemporal axis of the hippocampus and implementing spatial decoding analysis to describe the relationship between propagation and hippocampal replay.

Ph.D. Student — Laboratory for Neural Statistics, New York University, Flatiron Institute
May 2026 – Present

- Developed the mathematics and implementation for a novel statistical model for time-warping neural sequence detection.

Undergraduate Research Assistant — Smear Lab, University of Oregon
February 2023 – August 2025

- First authorship publication investigating relationships between olfactory bulb local field potentials and behavior using time-frequency analyses, signal processing approaches and predictive modeling. Developed novel statistical methods, performed all analyses, and wrote the manuscript.
- Co-authored publication in which I conducted spatial decoding analyses and developed the video-tracking pipeline.
- Co-launched project studying sniffing in Shank-3 mice where I performed implant surgeries and wrote analysis pipeline.
- Co-launched project studying behavior and neuronal dynamics in Anosmic mice. Worked with tetrodes and conducted analyses.

SKILLS

- **Computational**
 - Python (NumPy, Pandas, SciPy, PyTorch, JAX)
 - MATLAB (CellExplorer, Buzcode)
 - Hidden Markov models and state-space models
 - Neural decoding and Bayesian inference
 - Signal processing and time-frequency analysis
 - Statistical and dynamical systems modeling
 - Machine learning (SVMs, neural networks)
 - Spike-sorting (Kilosort, Phy, CellExplorer, custom scripts)
 - **Experimental**
 - Silicon probe implantation
 - Neuropixels recordings
 - Histology
 - Electrophysiology (Intan technologies, Open Ephys, Neuropixels)
 - Respiratory recordings
 - Video tracking (SLEAP, DeepLabCut)
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TEACHING

Teaching Assistant – Introduction to Computational Neuroscience, New York University
September 2026 – January 2027

Course Grader – Dynamical Systems and Control, University of Oregon
September 2024 – December 2024

AWARDS & HONORS

- **Training Program in Computational Neuroscience** – *New York University, 2026*
- **MacCracken Fellowship** – *New York University, 2025*
- **Undergraduate Research Award** — *Association for Chemoreception Sciences Conference, 2025*
- **Undergraduate Research Award** — *Association for Chemoreception Sciences Conference, 2024*
- **Apex Scholarship** — *University of Oregon, 2020*